# DATABASE SYSTEMS

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# NO DATABASE SYSTEMS TUTORIAL CLASS IN REMOTE LEARNING

- We will offer the class “database systens tutorial” only when you enter the NTUST campus

# TEXTBOOK

- Elmasri and Navathe "Fundamentals of Database Systems ", Seven Edition

Pearson, 2016

- Class Note

# GRADING POLICY

- Exams: There will be an in-class midterm (35 %) and a final exam (35 %)
- **Homework and Projects (30%):** SQLite and A WWW database based on MySQL, PHP

- Schedule Midterm, Final Exam and Homework and Project Submission after you enter Taiwan

# CLASS RULES

- No Make-up Tests:

Students are expected to take mid-term and final exams. We will not approve of absence from the midterm exam or final exam for any reason. If you do not take the mid-term exam or final exam, your final score is zero.

- Hand in your homework on time.
  - A homework submission with one week late is given 80% partial credit
  - Incomplete wok after a week is given a zero grade

# COURSE OUTLINE (BASIC ISSUES)

- Basic Database Concepts and Architecture
- Entity Relationship (E-R)Model and Enhanced Entity Relationship (EER) Model
- Relational Data Model, Query Algebra
- Structured Query Language (SQL)
- Database Design and Normalization

# COURSE OUTLINE (ADVNCED ISSUES)

- Transaction Management,
- Concurrency Control and Recovery
- Data warehousing
- Practical Database in Enterprise

# TYPES OF DATABASES AND DATABASE APPLICATIONS

- Traditional Applications:
  - Numeric and Textual Databases
- More Recent Applications:
  - Multimedia Databases
  - Geographic Information Systems (GIS)
  - Biological and Genome Databases
  - Data Warehouses
  - Mobile databases
  - Real-time and Active Databases

# RECENT DEVELOPMENTS (1)

- Social Networks started capturing a lot of information about people and about communications among people-posts, tweets, photos, videos in systems such as:

- Facebook

- Twitter

- Linked-In

- All of the above constitutes data
- Search Engines- Google, Bing, Yahoo : collect their own repository of web pages for searching purposes

# RECENT DEVELOPMENTS (2)

- New Technologies are emerging from the so-called non-database software vendors to manage vast amounts of data generated on the web:

- Big Data storage systems involving large clusters of distributed computers
- NOSQL (Not Only SQL) systems
- A large amount of data now resides on the "cloud" which means it is in huge data centers using thousands of machines.

# SIMPLIFIED DATABASE SYSTEM ENVIRONMENT

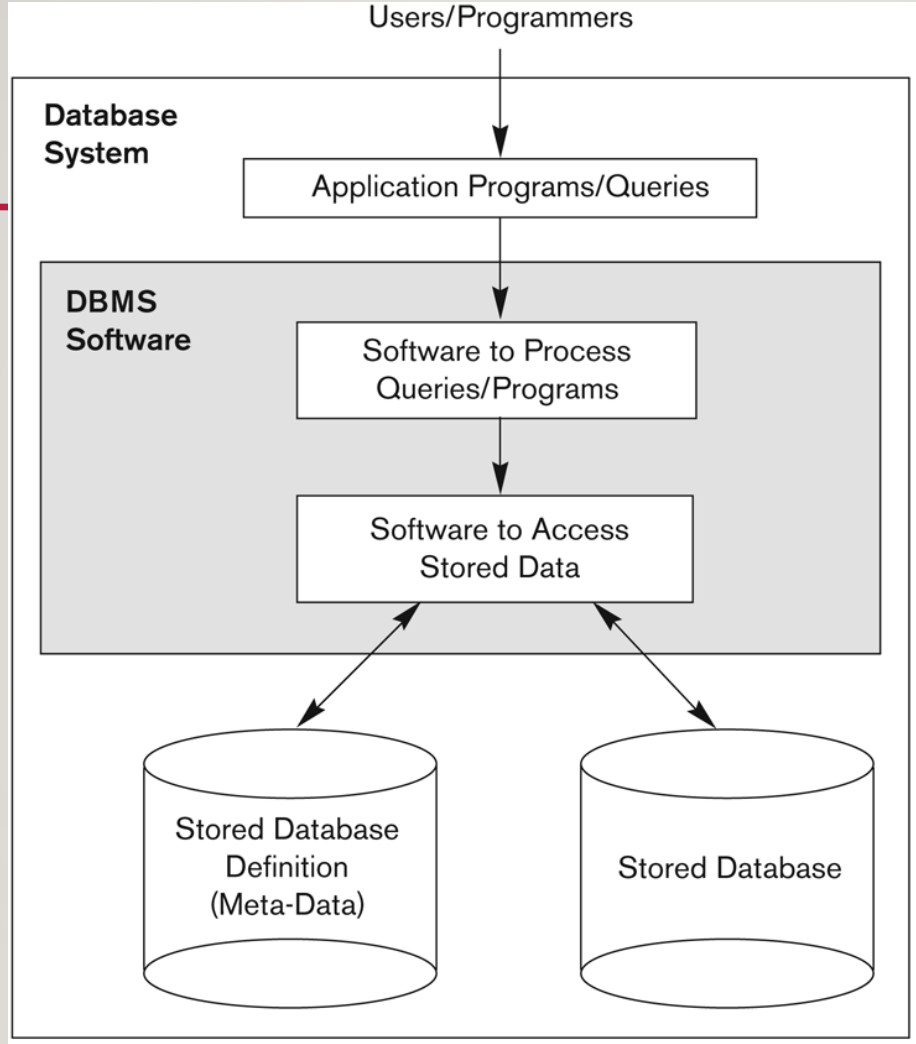
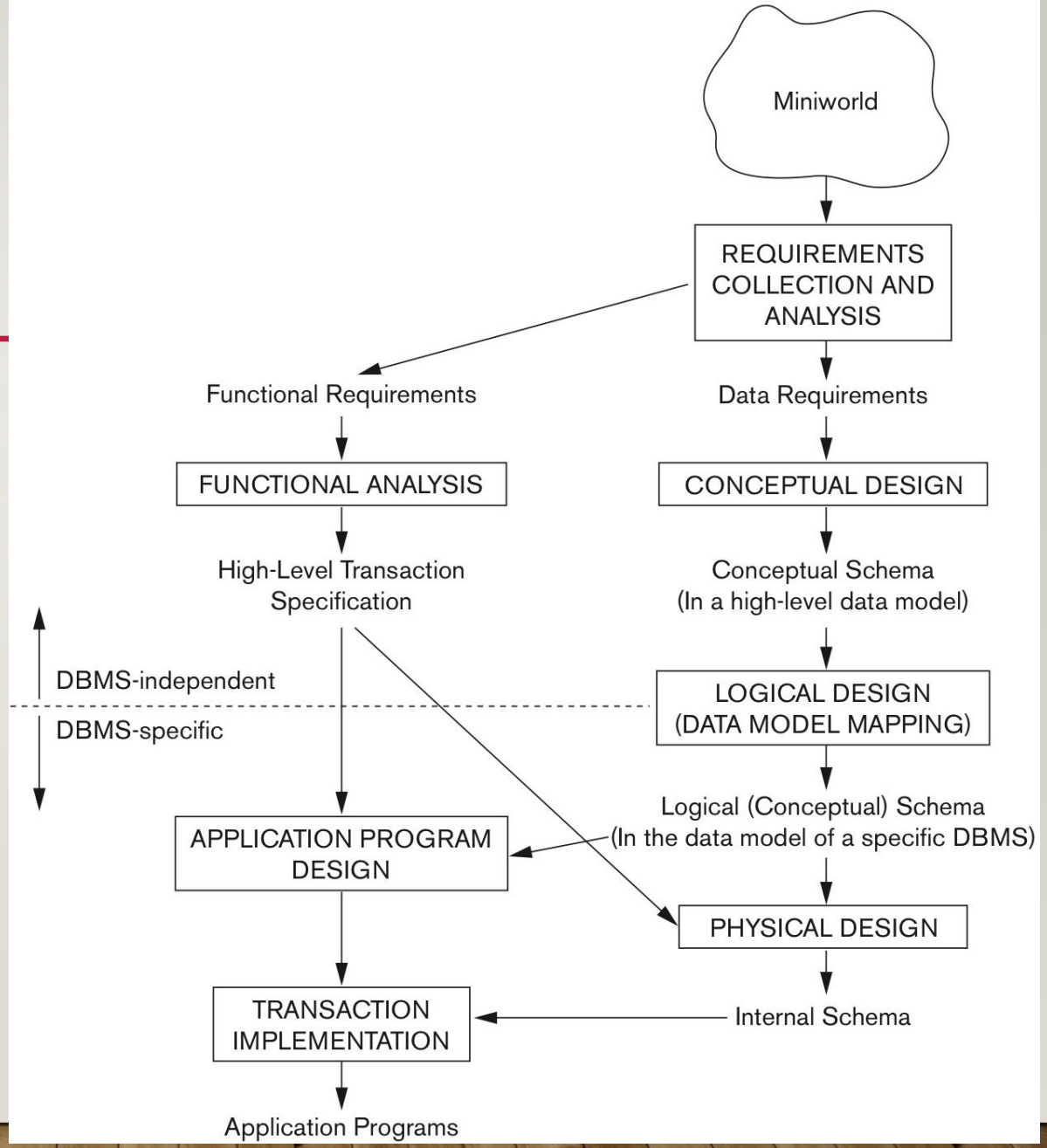



**Figure 1.1**
A simplified database system environment.

# MAIN PHASES OF DATABASE DESIGN



# EXAMPLE COMPANY DATABASE

- We need to create a database schema design based on the following (simplified) **requirements** of the COMPANY Database:
  - The company is organized into DEPARTMENTs. Each department has a name, number and an employee who *manages* the department. We keep track of the start date of the department manager. A department may have several locations.
  - Each department *controls* a number of PROJECTs. Each project has a unique name, unique number and is located at a single location.

# EXAMPLE COMPANY DATABASE (CONTINUED)

- The database will store each EMPLOYEE's social security number, address, salary, sex, and birthdate.
  - Each employee *works for* one department but may *work on* several projects.
  - The DB will keep track of the number of hours per week that an employee currently works on each project.
  - It is required to keep track of the *direct supervisor* of each employee.
- Each employee may *have* a number of DEPENDENTs.
  - For each dependent, the DB keeps a record of name, sex, birthdate, and relationship to the employee.

# ENTITY RELATIONSHIP DIAGRAM

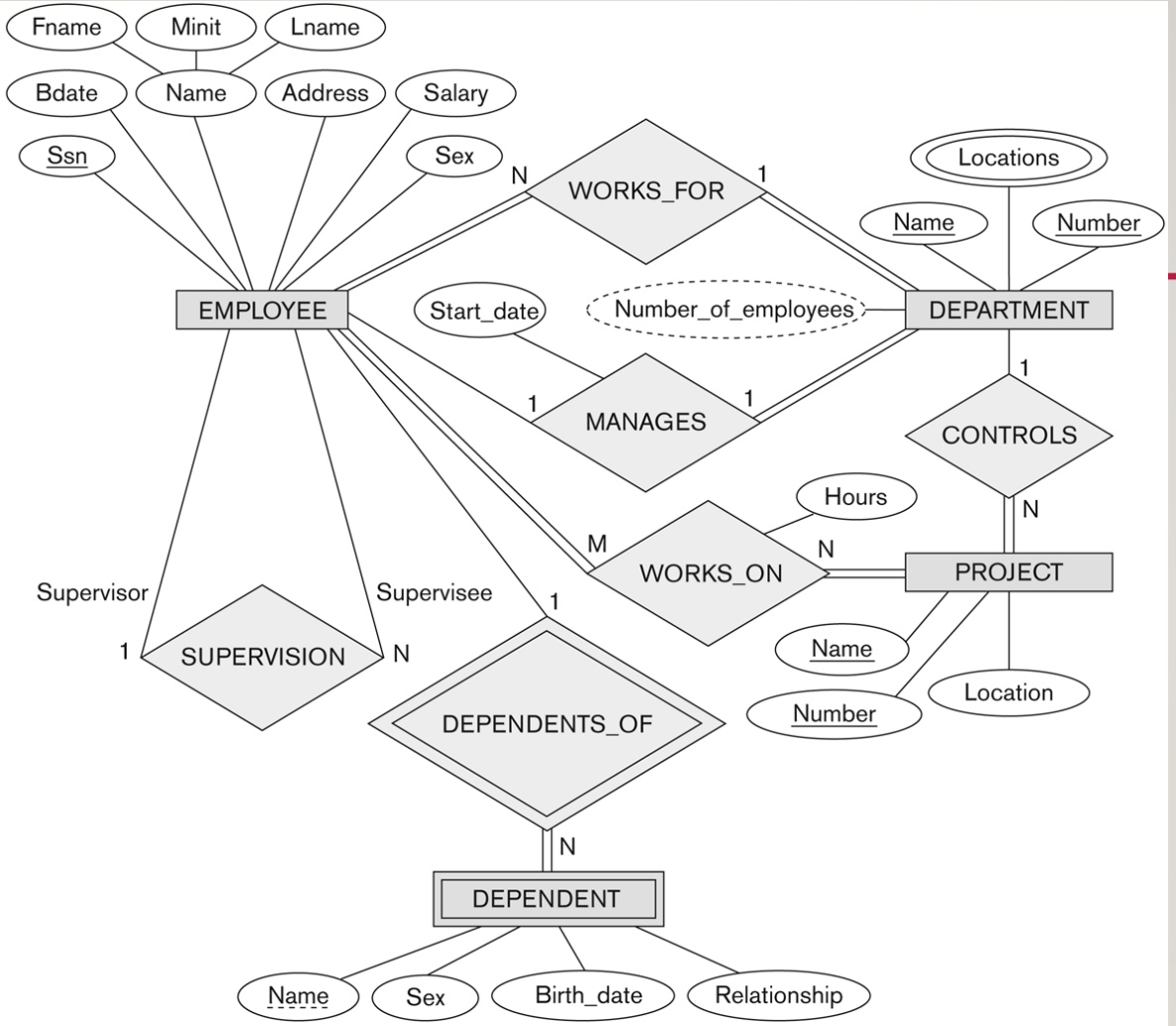



**Figure 3.2**
An ER schema diagram for the COMPANY database. The diagrammatic notation is introduced gradually throughout this chapter.

# EXAMPLE OF A SIMPLE RELATIONAL DATABASE

**COURSE**

| Course_name | Course_number | Credit_hours | Department |
|---|---|---|---|
| Intro to Computer Science | CS1310 | 4 | CS |
| Data Structures | CS3320 | 4 | CS |
| Discrete Mathematics | MATH2410 | 3 | MATH |
| Database | CS3380 | 3 | CS |

**SECTION**

| Section_identifier | Course_number | Semester | Year | Instructor |
|---|---|---|---|---|
| 85 | MATH2410 | Fall | 04 | King |
| 92 | CS1310 | Fall | 04 | Anderson |
| 102 | CS3320 | Spring | 05 | Knuth |
| 112 | MATH2410 | Fall | 05 | Chang |
| 119 | CS1310 | Fall | 05 | Anderson |
| 135 | CS3380 | Fall | 05 | Stone |

**GRADE_REPORT**

| Student_number | Section_identifier | Grade |
|---|---|---|
| 17 | 112 | B |
| 17 | 119 | C |
| 8 | 85 | A |
| 8 | 92 | A |
| 8 | 102 | B |
| 8 | 135 | A |

**PREREQUISITE**

| Course_number | Prerequisite_number |
|---|---|
| CS3380 | CS3320 |
| CS3380 | MATH2410 |
| CS3320 | CS1310 |

**Figure 1.2**
A database that stores student and course information.